Pricing conditions and comparative statics

Notation and definitions

We study one product and suppress its fixed index j: p_{jt} becomes p_t and s_{jt} becomes s_t. The outside-option share s_{0t} is written as $1 - s_t$. v_t is systematic component of utility.

Symbol	Definition
$t \in \{1, 2\}$	Time period.
j	Fixed product index, suppressed in displayed formulas.
p_t, p_t^*	Price and optimal price in period t, respectively.
$c \geq 0$	Constant marginal cost, common to both periods.
v_t	Systematic utility from purchasing in period t; outside utility is zero.
$s_t \in (0, 1)$	Logit purchase probability. In period 2 this is conditional on a period-1 purchase.
$1 - s_t$	Probability of choosing the outside option in period t.
$\alpha < 0$	Price coefficient in utility; $-\alpha > 0$ is price sensitivity.
β	Baseline utility or demand intercept.
γ	Utility coefficient on the history indicator, called switching preference in the figures.
$z_t \in \{0, 1\}$	Fixed, observed history indicator that shifts period-t utility by γz_t.
$\delta \in (0, 1)$	Discount factor on period-2 profit.
Π, Π^*	Expected discounted profit per initial customer, and its optimized value.
F_1, F_2	Period-1 profit derivative and normalized period-2 profit derivative; their roots give the pricing FOCs.
R^*	Optimal price ratio p_1^*/p_2^*.
∂, d	Partial derivatives hold the other price fixed; total derivatives allow both optimal prices and shares to adjust.

An asterisk denotes evaluation at the optimum. In all derivative formulas, unstarred prices and shares on the right-hand side are evaluated at that optimum.

$$\alpha = -2, \quad \beta = -1, \quad \delta = 0.7, \quad c = 1, \quad (z_1, z_2) \in \{(0,0), (1,0), (0,1)\}, \quad \gamma \in [0, 3].$$

Model and pricing conditions

$$v_1 = \alpha p_1 + \beta + \gamma z_1$$

$$v_2 = \alpha p_2 + \beta + \gamma z_2$$

$$s_1 = \frac{e^{\alpha p_1 + \beta + \gamma z_1}}{e^{\alpha p_1 + \beta + \gamma z_1} + 1}$$

$$s_2 = \frac{e^{\alpha p_2 + \beta + \gamma z_2}}{e^{\alpha p_2 + \beta + \gamma z_2} + 1}$$

$$\Pi = \delta s_1 s_2 \left(-c + p_2\right) + s_1 \left(-c + p_1\right)$$

First-order conditions for both prices

$$F_1 = \frac{\partial \Pi}{\partial p_1} = \alpha s_1 \left(1 - s_1\right) \left(-c + \delta s_2 \left(-c + p_2\right) + p_1\right) + s_1$$

$$\frac{\partial \Pi}{\partial p_2} = \delta s_1 F_2 = \delta s_1 \left(-\alpha c s_2 \left(1 - s_2\right) + \alpha p_2 s_2 \left(1 - s_2\right) + s_2\right)$$

$$F_2 = -\alpha c s_2 \left(1 - s_2\right) + \alpha p_2 s_2 \left(1 - s_2\right) + s_2$$

$$F_1 = F_2 = 0$$

$$0 = \frac{F_1}{s_1} = -\alpha \left(1 - s_1\right) \left(c + \delta s_2 \left(c - p_2\right) - p_1\right) + 1$$

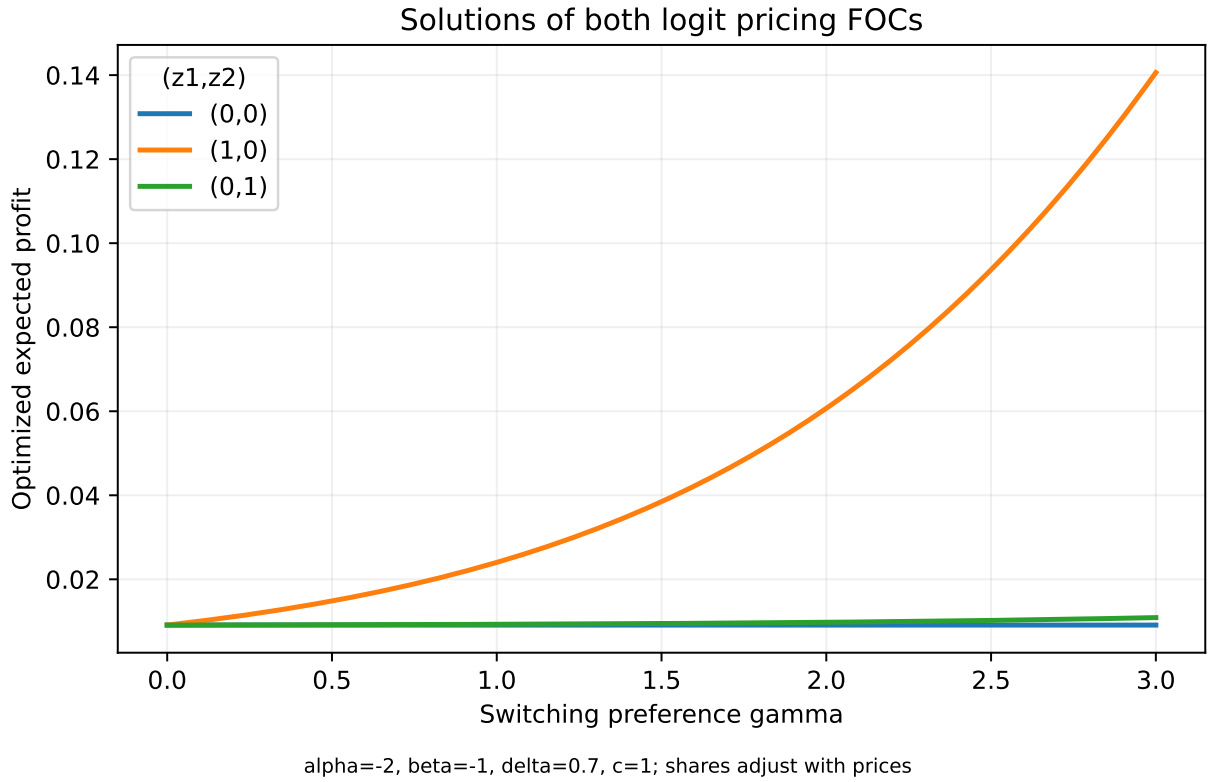
$$0 = \frac{F_2}{s_2} = -\alpha c \left(1 - s_2\right) + \alpha p_2 \left(1 - s_2\right) + 1$$

$$p_2^* = c - \frac{1}{\alpha \left(1 - s_2\right)}$$

$$p_1^* = c + \frac{\delta s_2}{\alpha \left(1 - s_2\right)} - \frac{1}{\alpha \left(1 - s_1\right)}$$

Optimized expected profit

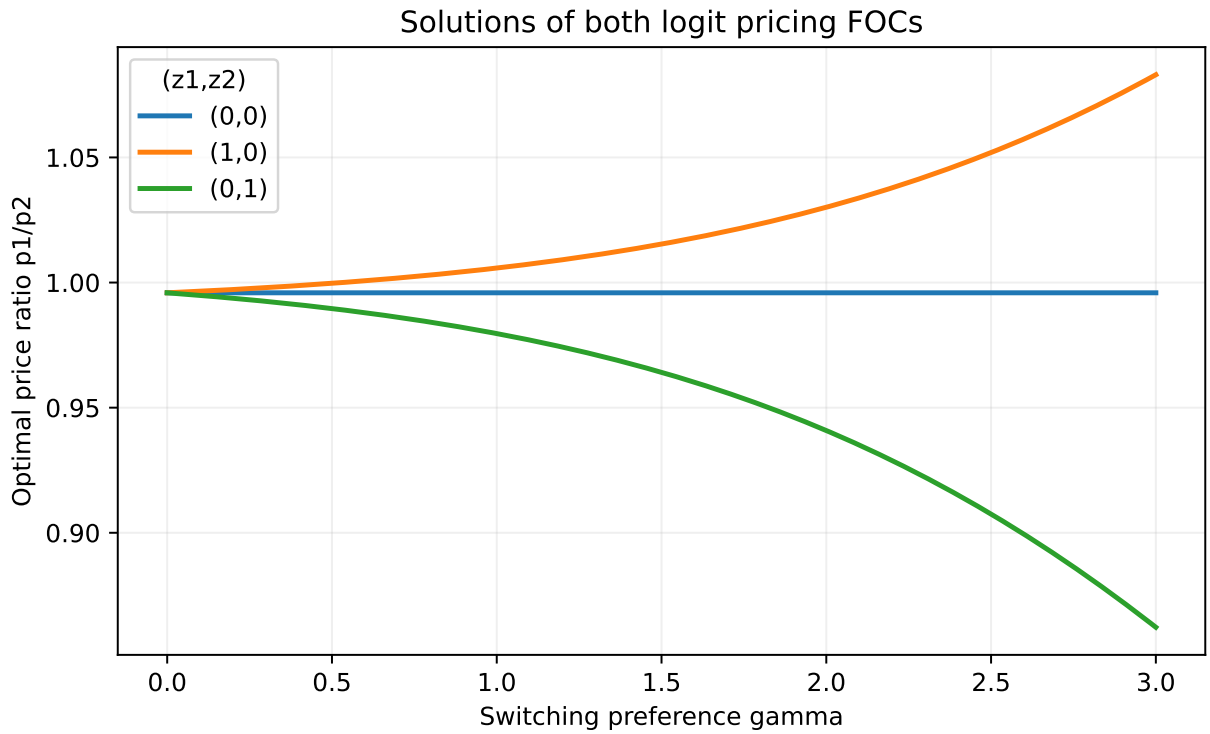
$$\Pi^* = \delta s_1 s_2 \left(-c + p_2\right) + s_1 \left(-c + p_1\right)$$



alpha=-2, beta=-1, delta=0.7, c=1; shares adjust with prices

Optimal price ratio

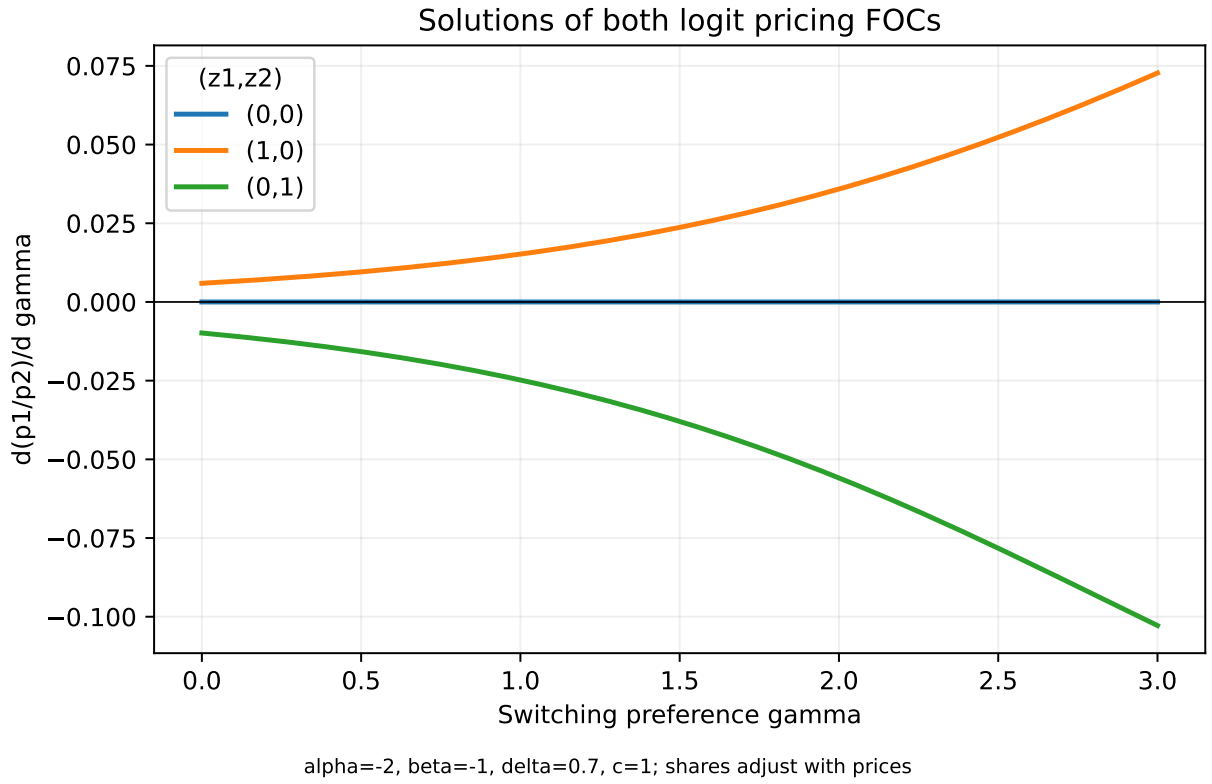
$$R^* = \frac{p_1}{p_2}$$



alpha=-2, beta=-1, delta=0.7, c=1; shares adjust with prices

Price-ratio derivative in gamma

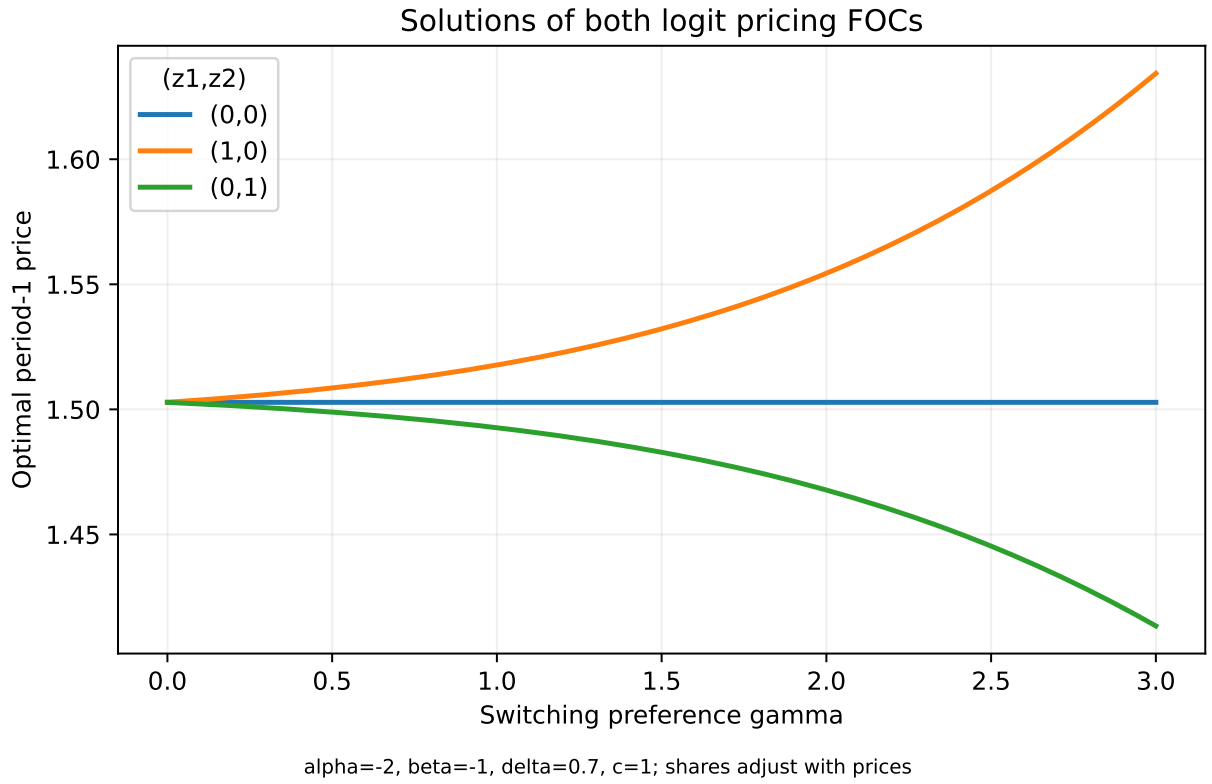
$$\frac{\mathrm{d}R^*}{\mathrm{d}\gamma} = \frac{\delta p_2 s_2 z_2 \left(1 - s_1\right) + p_1 s_2 z_2 - p_2 s_1 z_1}{\alpha p_2^2}$$



alpha=-2, beta=-1, delta=0.7, c=1; shares adjust with prices

Optimal period-1 price

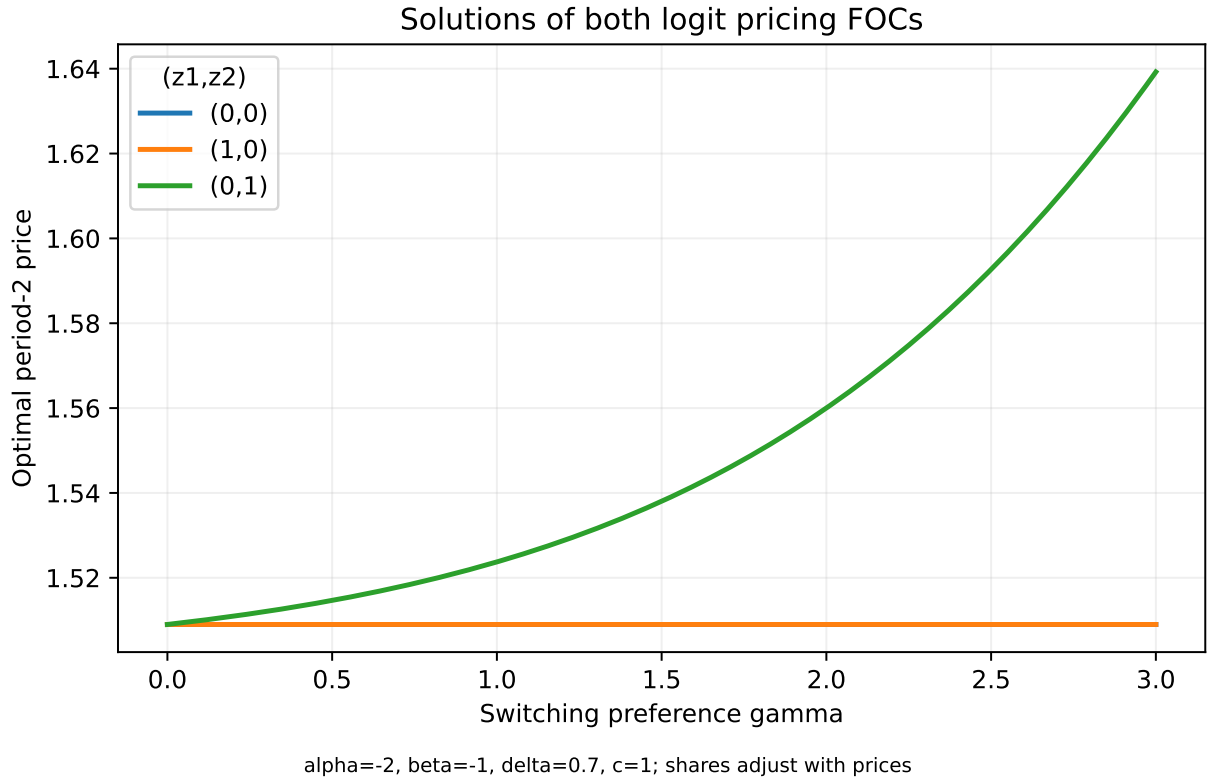
$$p_1^* = c + \frac{\delta s_2}{\alpha\,(1 - s_2)} - \frac{1}{\alpha\,(1 - s_1)}$$



alpha=-2, beta=-1, delta=0.7, c=1; shares adjust with prices

Optimal period-2 price

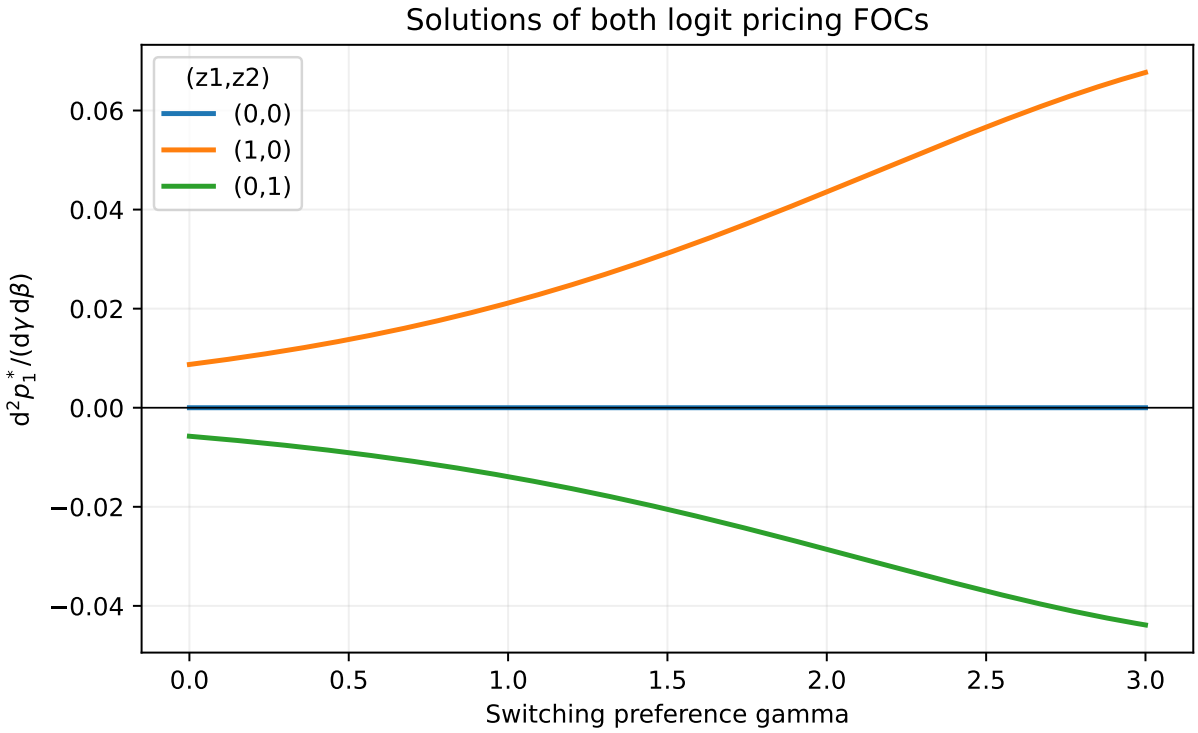
$$p_2^* = c - \frac{1}{\alpha\,(1 - s_2)}$$



alpha=-2, beta=-1, delta=0.7, c=1; shares adjust with prices

$$\mathrm{d}^2 p_1^* / (\mathrm{d}\gamma\, \mathrm{d}\beta)$$

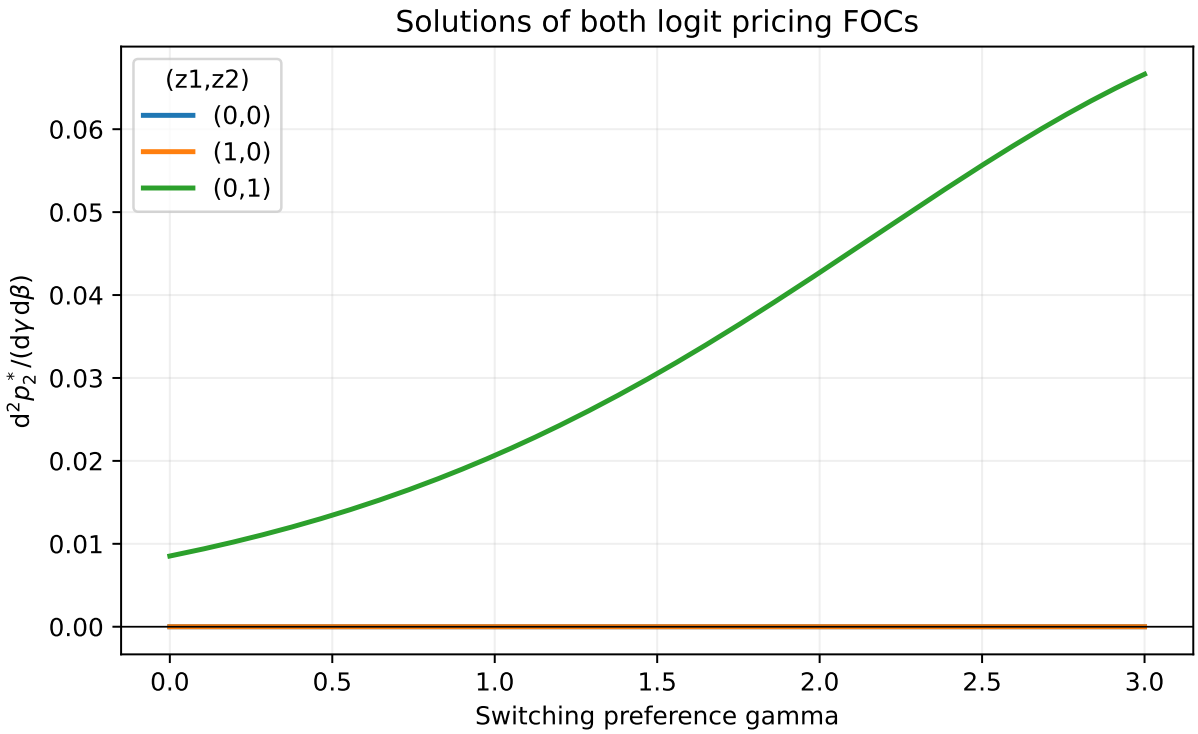
$$\mathrm{d}^2 p_1^* / (\mathrm{d}\gamma\, \mathrm{d}\beta) = -\frac{(1-s_1)\left(\delta^2 s_1 s_2^2 z_2 (1-s_1) - \delta s_1^2 s_2 z_1 + \delta s_1 s_2 z_1 + \delta s_1 s_2 z_2 (1-s_1) + \delta s_2^2 z_2 (1-s_2) - \delta s_2 z_2 (1-s_2) - s_1^2 z_1 + s_1 z_1\right)}{\alpha}$$



alpha=-2, beta=-1, delta=0.7, c=1; shares adjust with prices

$\mathrm{d}^2 p_2^* / (\mathrm{d}\gamma \, \mathrm{d}\beta)$

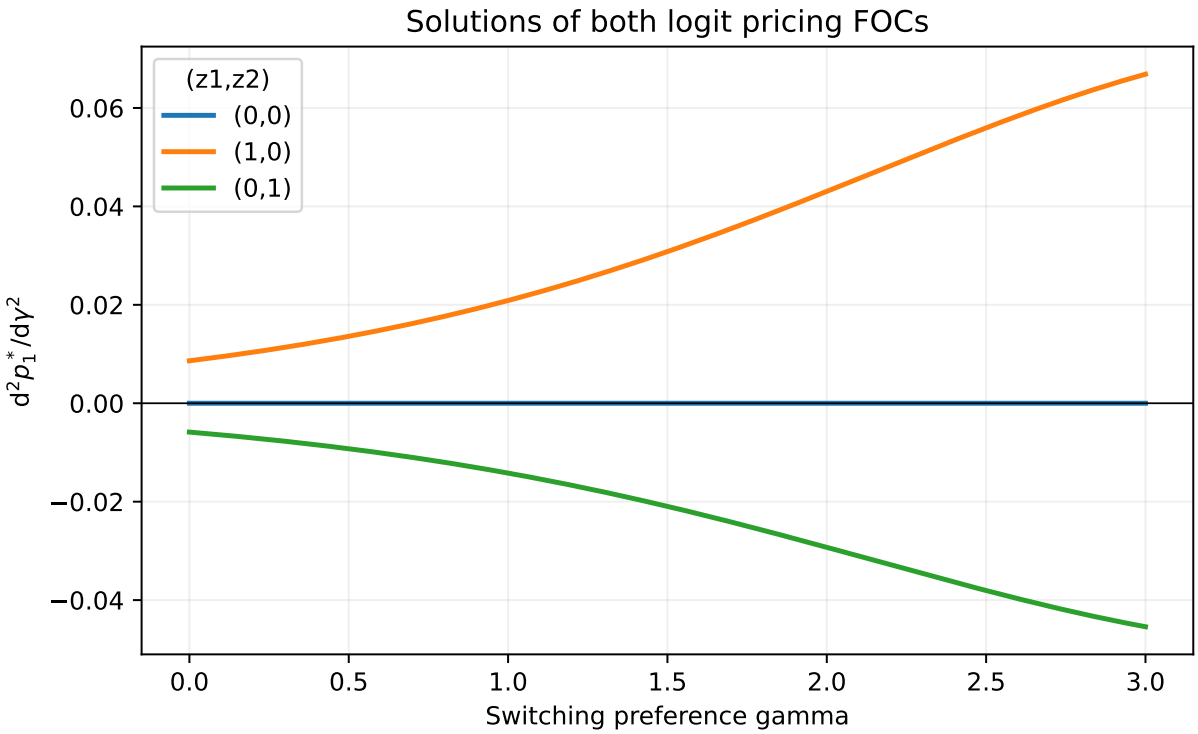
$$\mathrm{d}^2 p_2^* / (\mathrm{d}\gamma \, \mathrm{d}\beta) = -\frac{s_2 z_2 \left(1 - s_2\right)^2}{\alpha}$$



alpha=-2, beta=-1, delta=0.7, c=1; shares adjust with prices

$\mathrm{d}^2 p_1^*/\mathrm{d}\gamma^2$

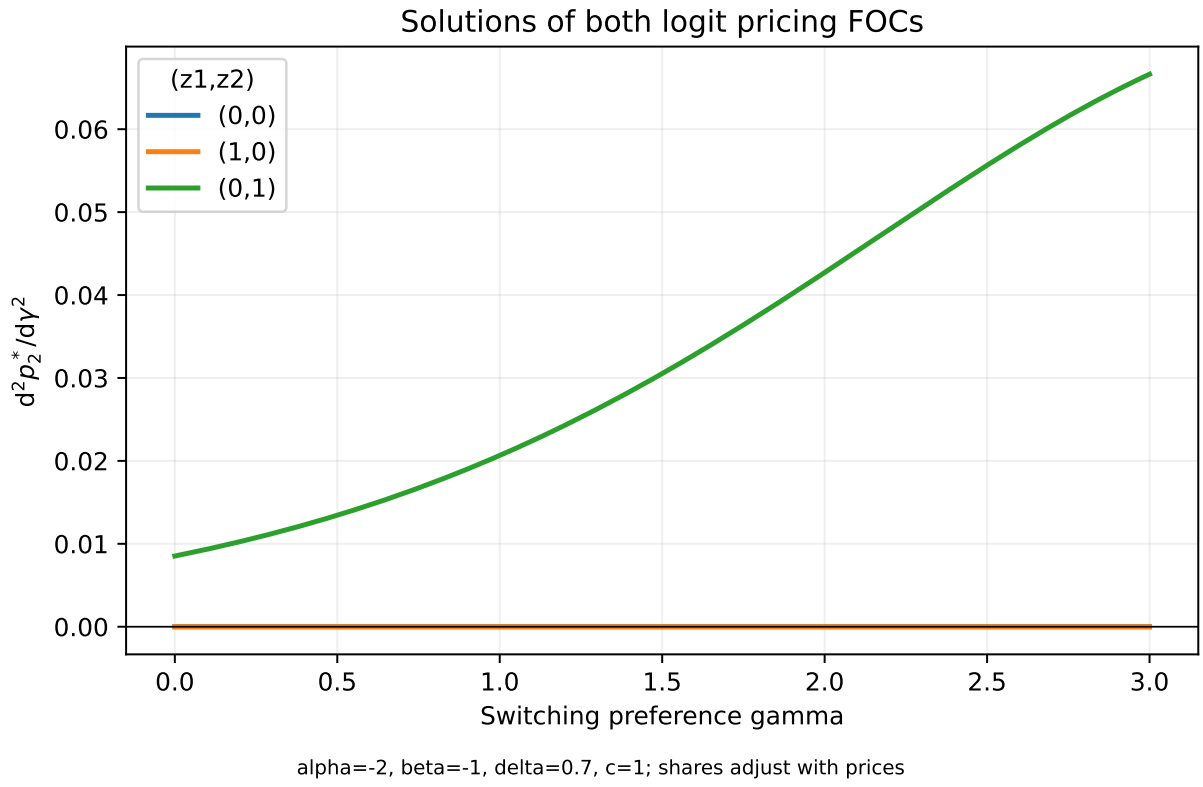
$$\mathrm{d}^2 p_1^*/\mathrm{d}\gamma^2 = \frac{\delta s_2 z_2^2 \left(1 - s_1\right)\left(1 - s_2\right)^2 - s_1 \left(1 - s_1\right)^2 \left(\delta s_2 z_2 + z_1\right)^2}{\alpha}$$



alpha=-2, beta=-1, delta=0.7, c=1; shares adjust with prices

$\mathrm{d}^2 p_2^*/\mathrm{d}\gamma^2$

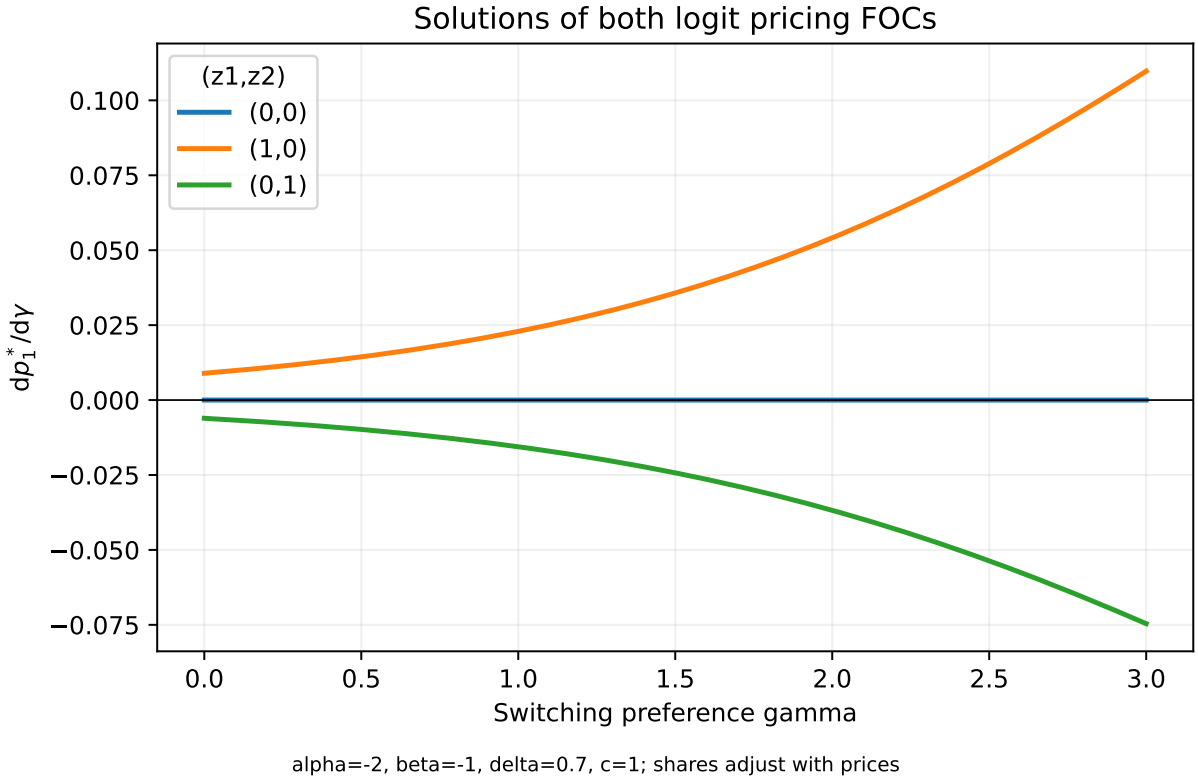
$$\mathrm{d}^2 p_2^*/\mathrm{d}\gamma^2 = -\frac{s_2 z_2^2 \left(1 - s_2\right)^2}{\alpha}$$



alpha=-2, beta=-1, delta=0.7, c=1; shares adjust with prices

$\mathrm{d}p_1^*/\mathrm{d}\gamma$

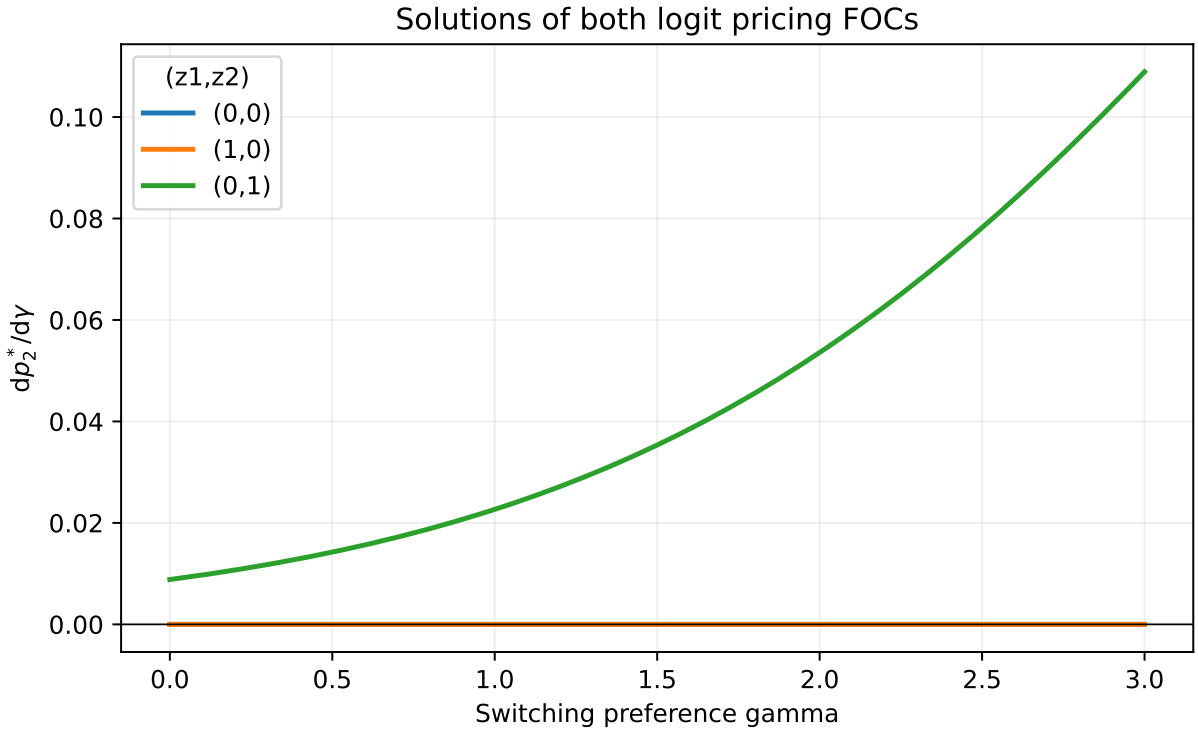
$$\mathrm{d}p_1^*/\mathrm{d}\gamma = \frac{\delta s_2 z_2 \left(1 - s_1\right) - s_1 z_1}{\alpha}$$



alpha=-2, beta=-1, delta=0.7, c=1; shares adjust with prices

$\mathrm{d}p_2^*/\mathrm{d}\gamma$

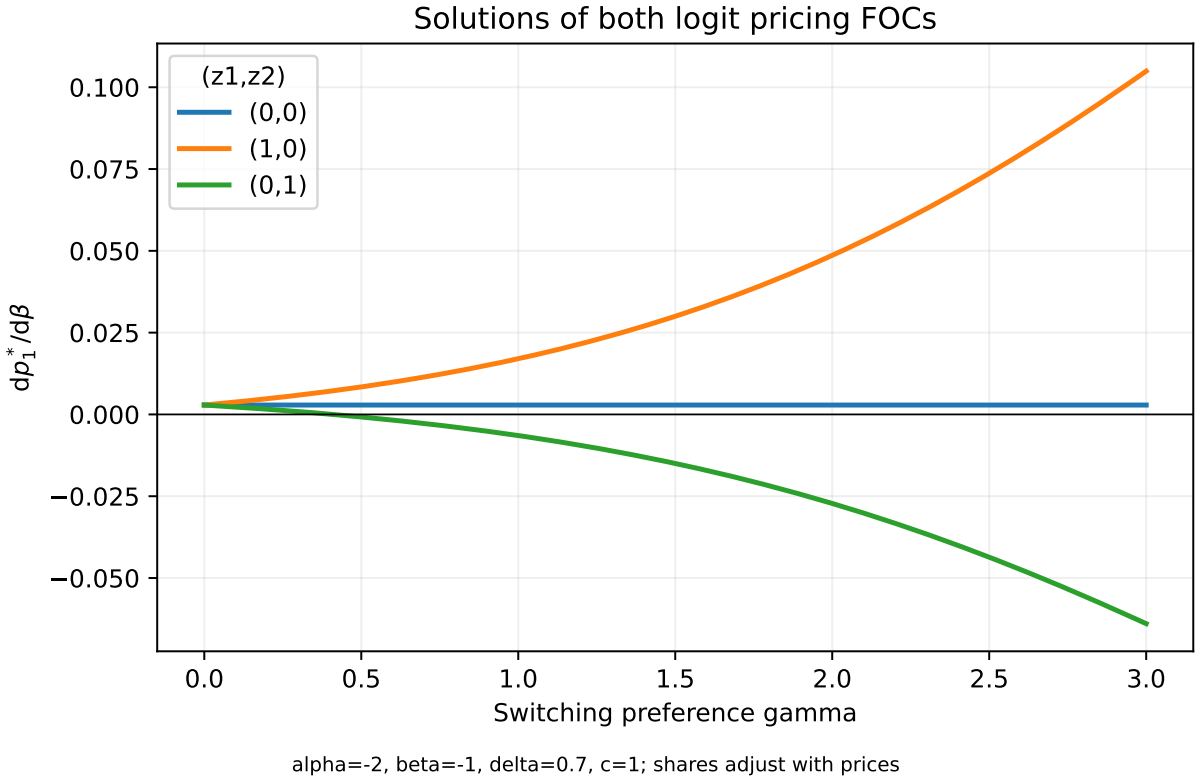
$$\mathrm{d}p_2^*/\mathrm{d}\gamma = -\frac{s_2 z_2}{\alpha}$$



alpha=-2, beta=-1, delta=0.7, c=1; shares adjust with prices

$\mathrm{d}p_1^*/\mathrm{d}\beta$

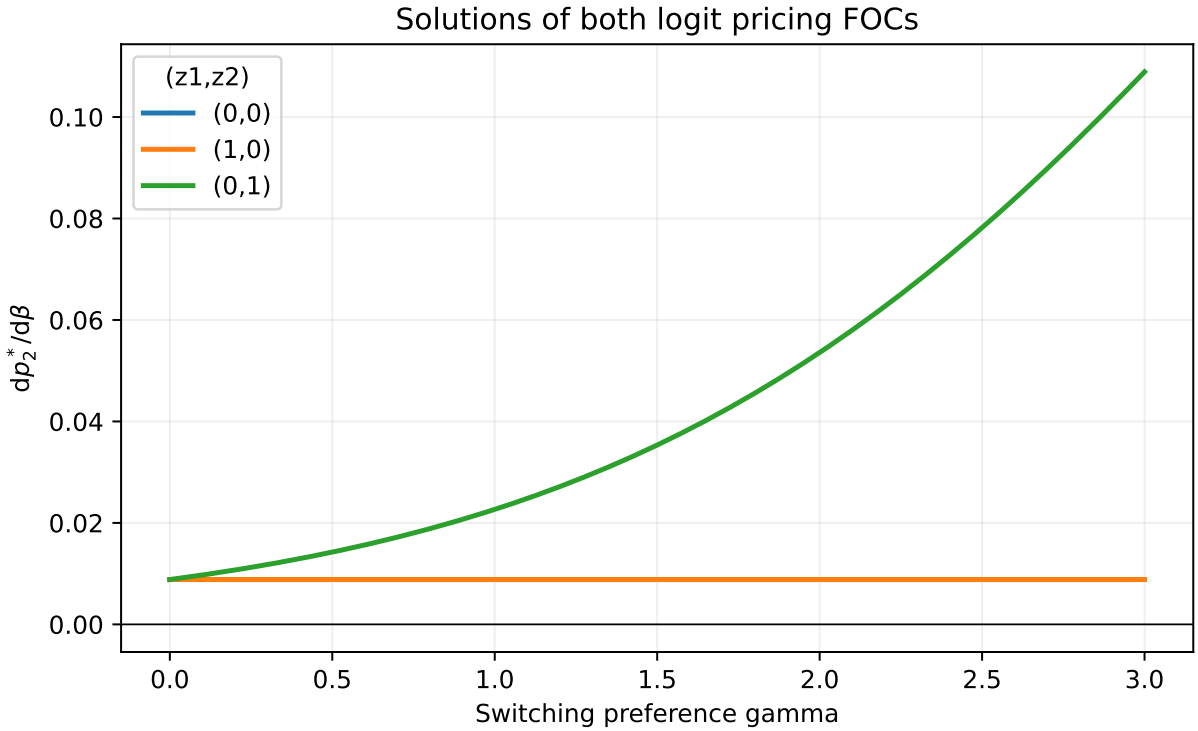
$$\mathrm{d}p_1^*/\mathrm{d}\beta = \frac{\delta s_2 \left(1 - s_1\right) - s_1}{\alpha}$$



alpha=-2, beta=-1, delta=0.7, c=1; shares adjust with prices

$\mathrm{d}p_2^*/\mathrm{d}\beta$

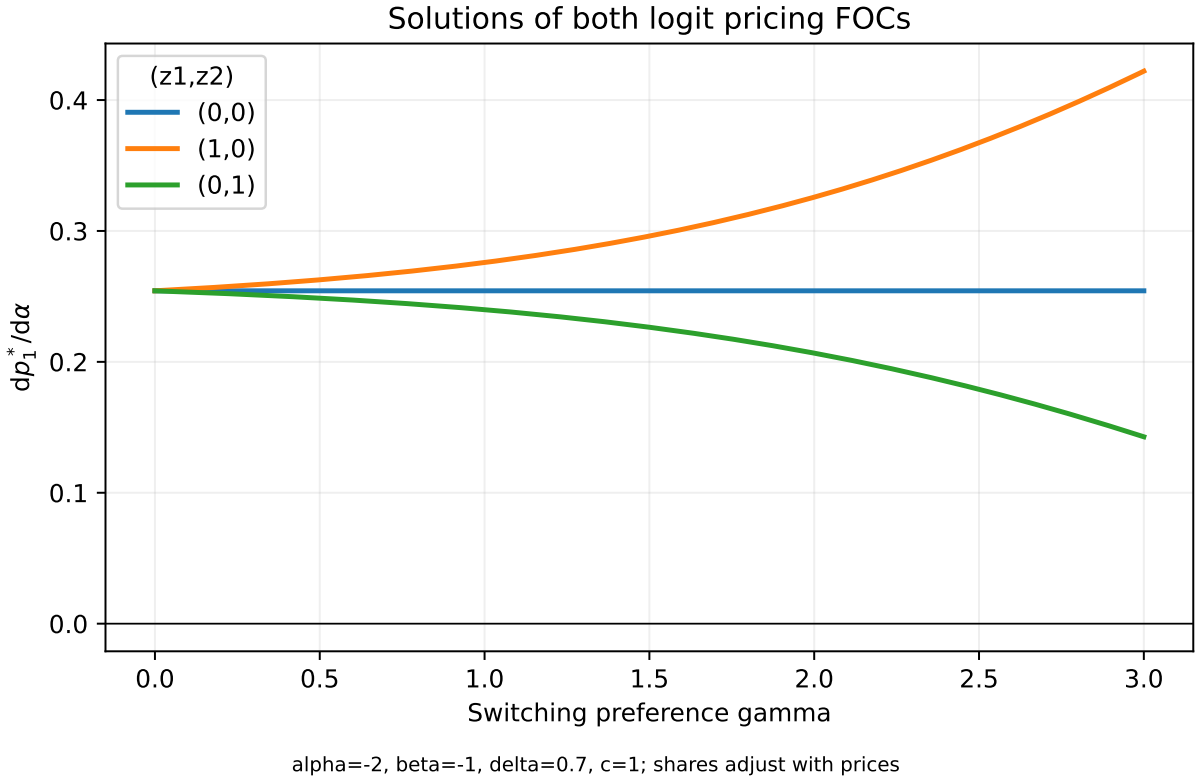
$$\mathrm{d}p_2^*/\mathrm{d}\beta = -\frac{s_2}{\alpha}$$



alpha=-2, beta=-1, delta=0.7, c=1; shares adjust with prices

$\mathrm{d}p_1^*/\mathrm{d}\alpha$

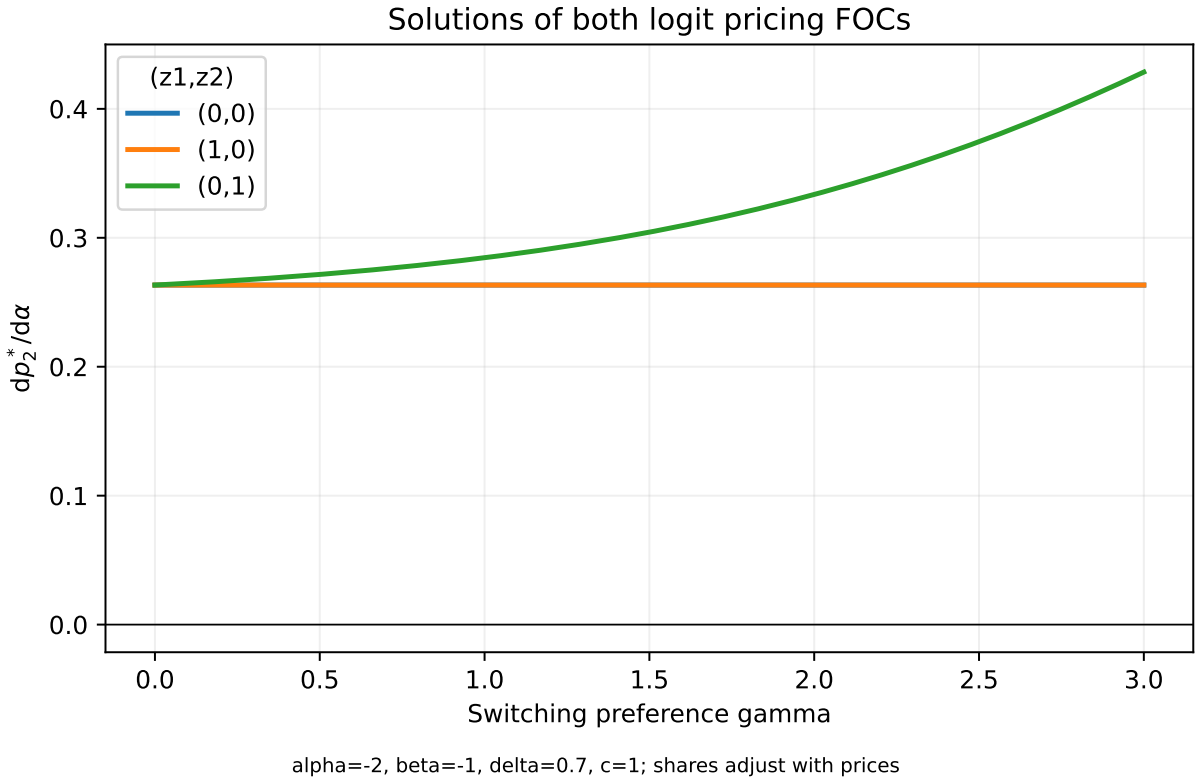
$$\mathrm{d}p_1^*/\mathrm{d}\alpha = \frac{\delta p_2 s_2 \left(1 - s_1\right)}{\alpha} - \frac{p_1 s_1}{\alpha} + \frac{1}{\alpha^2}$$



alpha=-2, beta=-1, delta=0.7, c=1; shares adjust with prices

$\mathrm{d}p_2^*/\mathrm{d}\alpha$

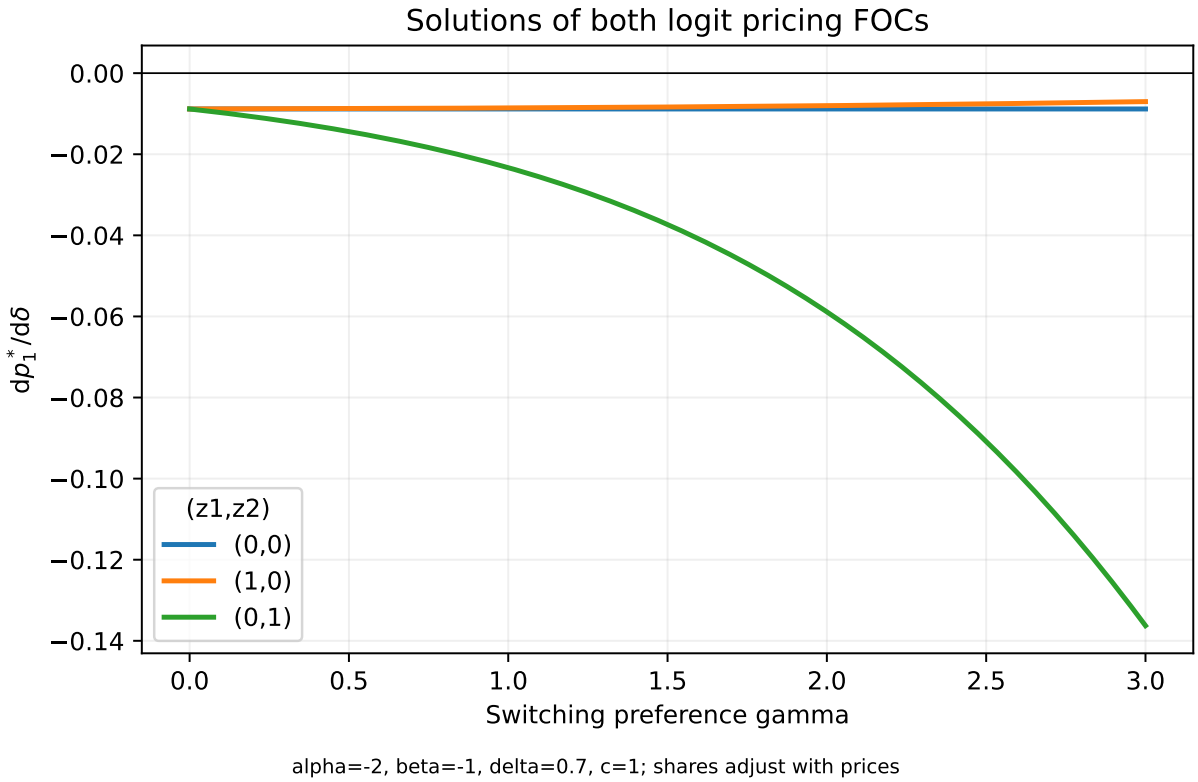
$$\mathrm{d}p_2^*/\mathrm{d}\alpha = \frac{-\alpha p_2 s_2 + 1}{\alpha^2}$$



alpha=-2, beta=-1, delta=0.7, c=1; shares adjust with prices

$\mathrm{d}p_1^*/\mathrm{d}\delta$

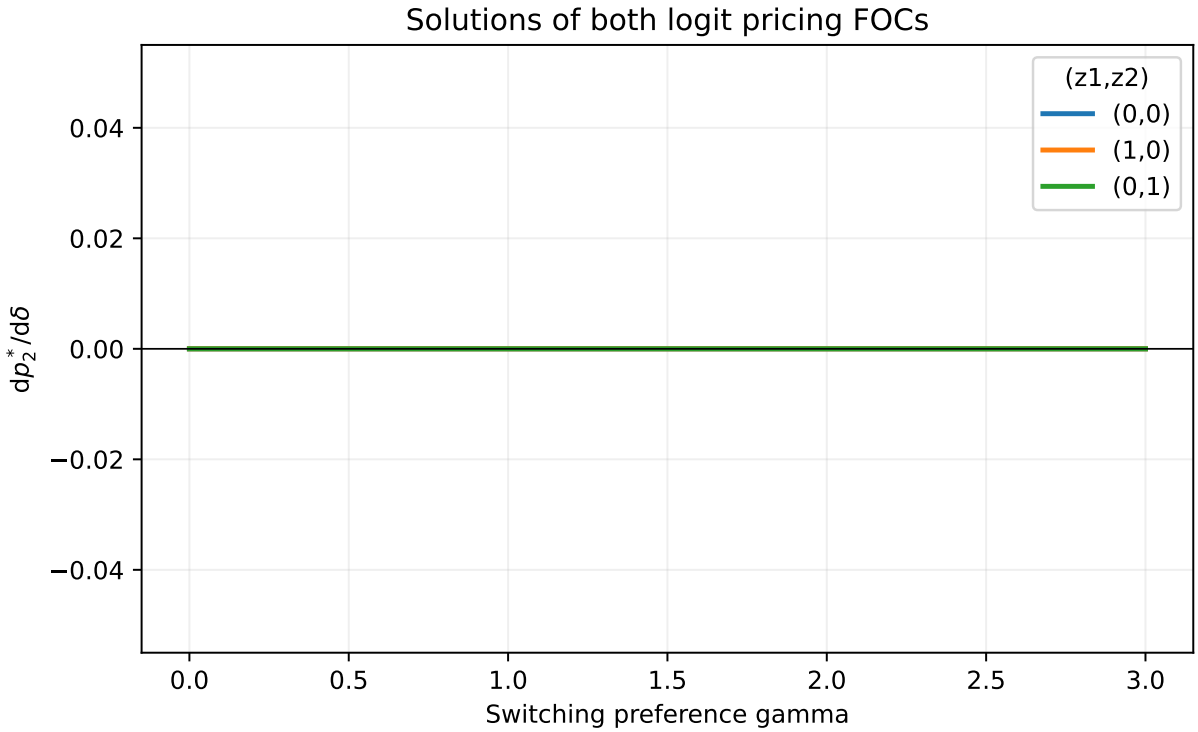
$$\mathrm{d}p_1^*/\mathrm{d}\delta = \frac{s_2\,(1-s_1)}{\alpha\,(1-s_2)}$$



alpha=-2, beta=-1, delta=0.7, c=1; shares adjust with prices

$\mathrm{d}p_2^*/\mathrm{d}\delta$

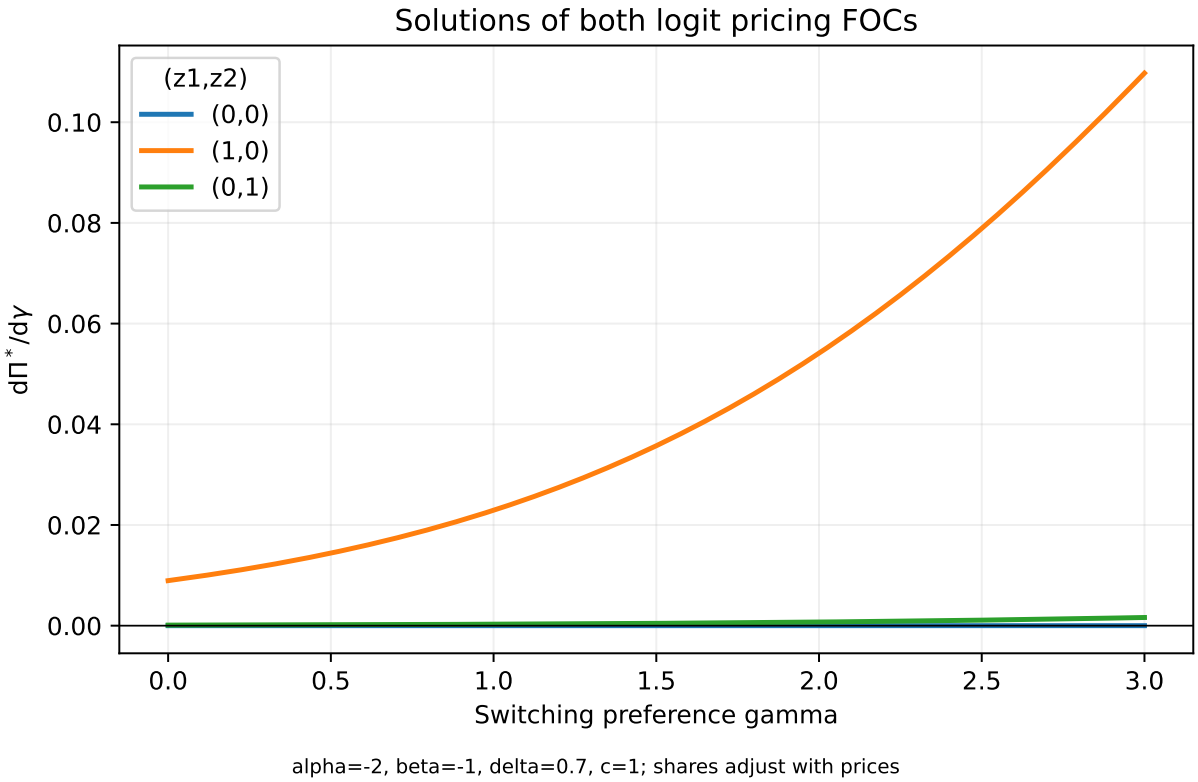
$$\mathrm{d}p_2^*/\mathrm{d}\delta = 0$$



alpha=-2, beta=-1, delta=0.7, c=1; shares adjust with prices

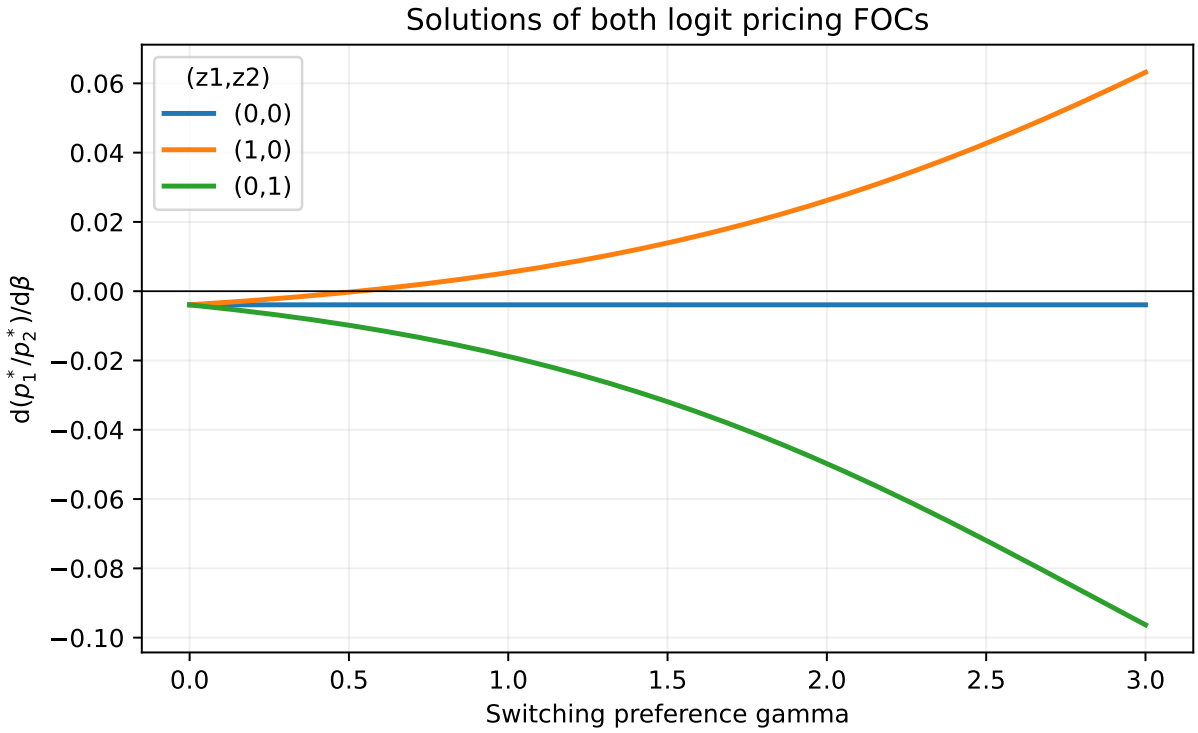
$\mathrm{d}\Pi^*/\mathrm{d}\gamma$

$$\mathrm{d}\Pi^*/\mathrm{d}\gamma = -\frac{s_1\left(\delta s_2 z_2 + z_1\right)}{\alpha}$$



$\mathrm{d}(p_1^*/p_2^*)/\mathrm{d}\beta$

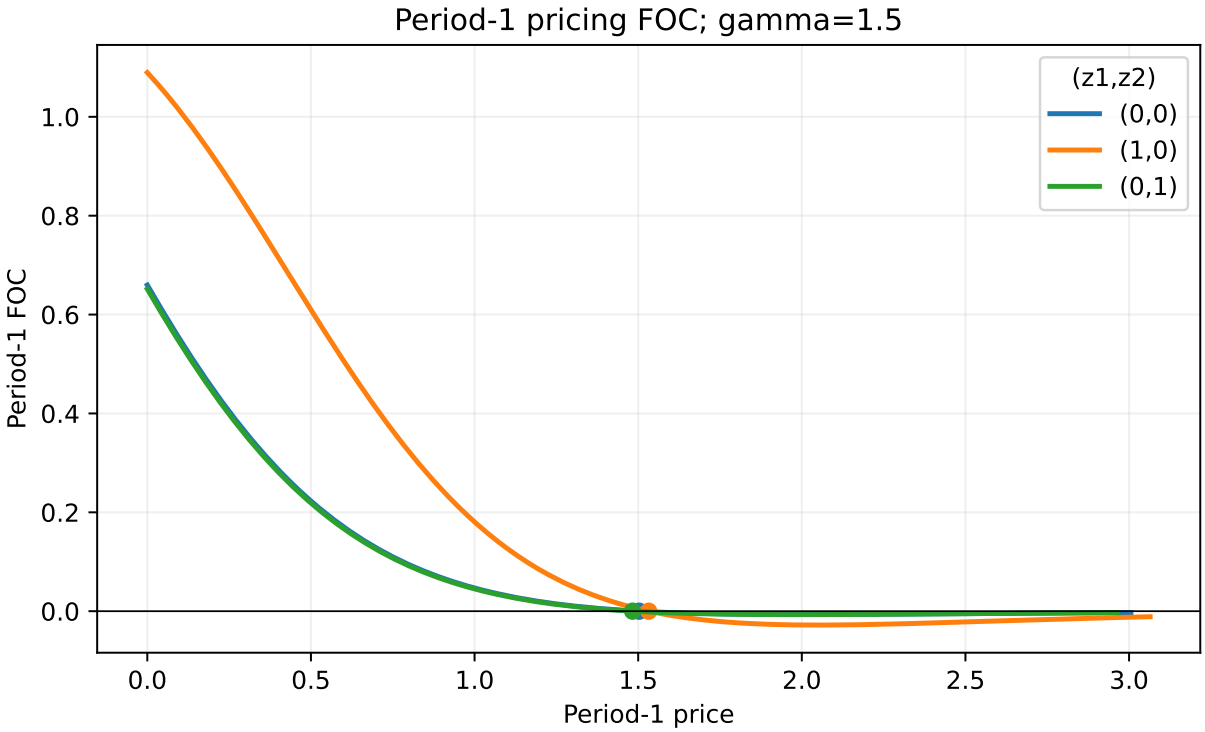
$$\mathrm{d}(p_1^*/p_2^*)/\mathrm{d}\beta = \frac{\frac{p_1 s_2}{\alpha} + \frac{p_2(\delta s_2(1-s_1)-s_1)}{\alpha}}{p_2^2}$$



alpha=-2, beta=-1, delta=0.7, c=1; shares adjust with prices

Period-1 pricing condition

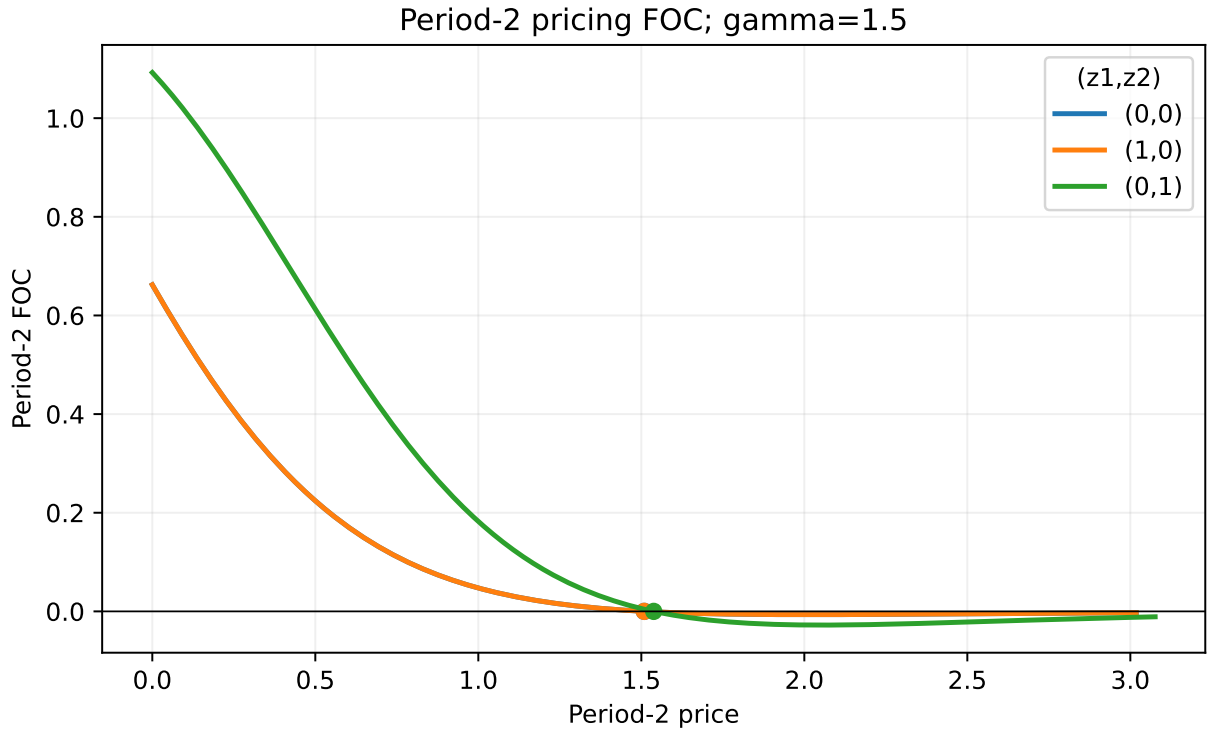
$$F_1 = \alpha s_1 (1 - s_1) (-c + \delta s_2 (-c + p_2) + p_1) + s_1$$



alpha=-2, beta=-1, delta=0.7, c=1; other price fixed at optimum

Period-2 pricing condition

$$F_2 = -\alpha c s_2 (1 - s_2) + \alpha p_2 s_2 (1 - s_2) + s_2$$



alpha=-2, beta=-1, delta=0.7, c=1; other price fixed at optimum